

Unit - IV

Bloch Theorem

Statement:- The Bloch theorem is a mathematical statement regarding the form of the solutions of the Schrodinger equation of an electron moving in a region of periodic potential. It states that the eigenfunctions of the electronic Schrodinger equation for a periodic potential are

$$\psi(\vec{r}) = e^{i\vec{k} \cdot \vec{r}} u_k(\vec{r}) \quad \text{--- (1)}$$

where $u_k(\vec{r})$ has the period of the crystal lattice. Eq. (1) is the statement of Bloch theorem which means that the eigenfunctions of the wave equation for a periodic potential are the form of plane waves $e^{i\vec{k} \cdot \vec{r}}$, modulated by a function $u_k(\vec{r})$ depends on the wave vector $\vec{k}$.

According to Bloch theorem, the solutions of one-dimensional Schrodinger equation are the free electron plane waves, $e^{\pm i k x}$, modulated by the function $u_k(x)$ which has the same periodicity 'a' as the lattice. The solutions

are

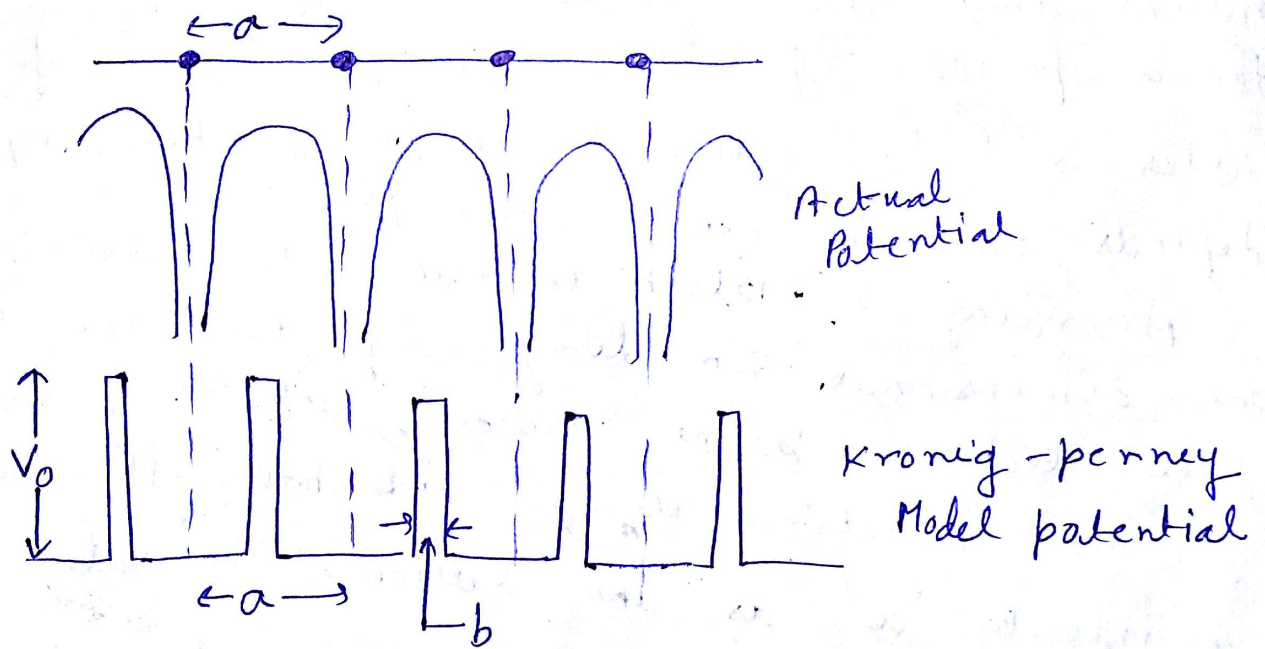
$$\psi(x) = e^{\pm i k x} u_k(x) \quad \text{--- (2)}$$

where

$$u_k(x) = u_k(x+a) \quad \text{--- (3)}$$

KRONIG - PENNEY MODEL:-

Kronig and Penney suggested a simplified model potential consisting of an infinite row of rectangular potential wells separated by barriers of width 'b' with space periodicity 'a' which is the periodicity of the lattice. In the region $0 < x < a$, the potential energy equals to zero and in the region $-b < x < 0$ it is V_0 . This model potential is shown in fig. below



The solution of Schrodinger equation for an electron in a periodic lattice is found by using Bloch theorem.

$$\psi(x) = u_k(x) e^{ikx}$$

where the modulating factor $u_k(x)$ is periodic with the same periodicity a as of the potential, i.e.,

$$\psi_k(x) = \psi_k(x+a)$$

$$u_k(x) = u_k(x+a) = u_k(x+na)$$

where n is an integer.

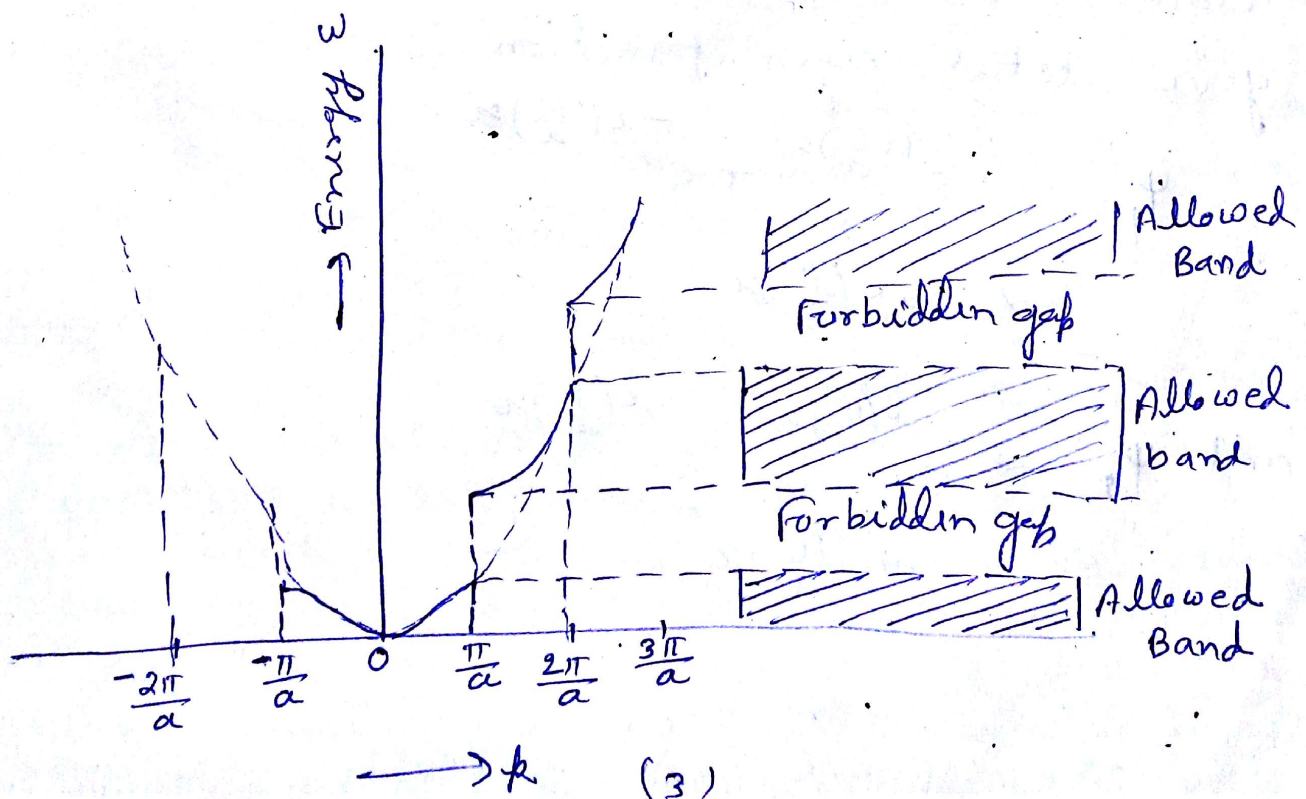
The wavefunction is $i(ka - \omega t)$

$$\psi(x, t) = u_k(x) e^{i(ka - \omega t)}$$

where exponential term describes a wave of wavelength $\lambda = 2\pi/k$, which travels toward $+x$ if k is $+ve$, and toward $-x$ if k is negative.

On solving Schrodinger equation of the electron for the $K-P$ potential under the condition that ψ and $d\psi/dx$ must be continuous at the boundaries of the well, the allowed energies in terms of k of the electron is obtained which shows that gaps in energy occur at values given by

$$k = \pm \frac{\pi}{a}, \pm \frac{2\pi}{a}, \pm \frac{3\pi}{a} \dots$$



The Bragg's condition for the reflection of waves is

$$2a \sin \theta = n\lambda \quad n = 1, 2, 3, \dots$$

where a is the spacing between the ions of the lattice. Since the lattice is one dimension only, the above equation becomes

$$2a = n\lambda$$

$$[\theta = 90^\circ]$$

$$\text{But } \lambda = \frac{2\pi}{k}$$

$$\therefore 2a = n \cdot \frac{2\pi}{k}$$

$$\text{or } k = n \frac{\pi}{a} = \pm \frac{\pi}{a}, \pm \frac{2\pi}{a}, \pm \frac{3\pi}{a}, \dots$$

The eigen functions describing the incident and the corresponding reflected waves are e^{ikx} or $e^{i(\frac{\pi}{a})x}$ and $e^{-i(\frac{\pi}{a})x}$. The two

eigen functions can combine in two types to give total eigen function

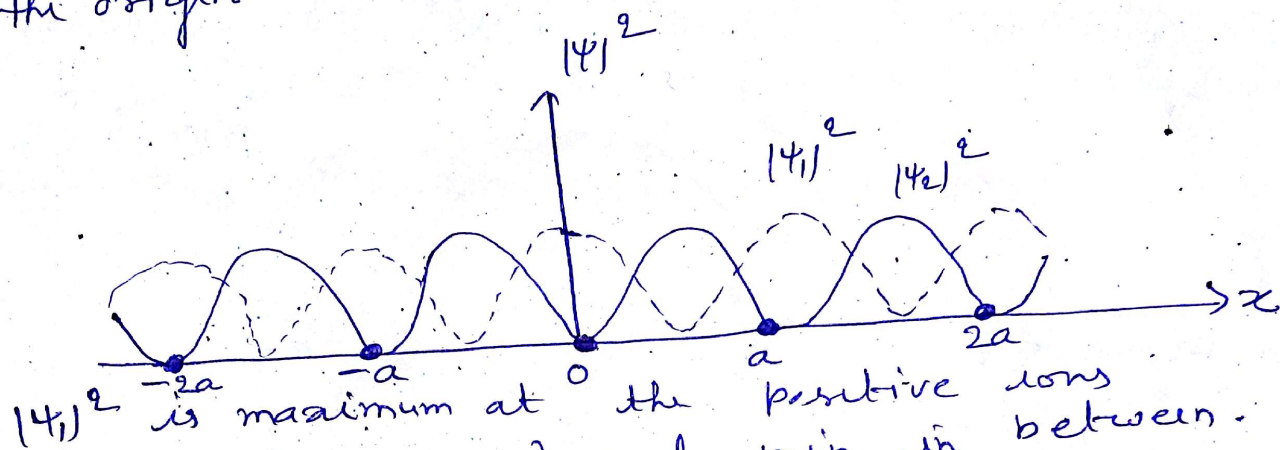
$$\psi_1 = e^{i(\frac{\pi}{a})x} + e^{-i(\frac{\pi}{a})x}$$

$$\propto \cos\left(\frac{\pi}{a}x\right)$$

$$\text{and } \psi_2 = e^{i(\frac{\pi}{a})x} - e^{-i(\frac{\pi}{a})x}$$

$$\propto \sin\left(\frac{\pi}{a}x\right)$$

Thus we get two different possible standing waves. The probability densities $|\psi_1|^2$ and $|\psi_2|^2$ for these waves are plotted in fig. in which the origin has been taken at one of the ions.



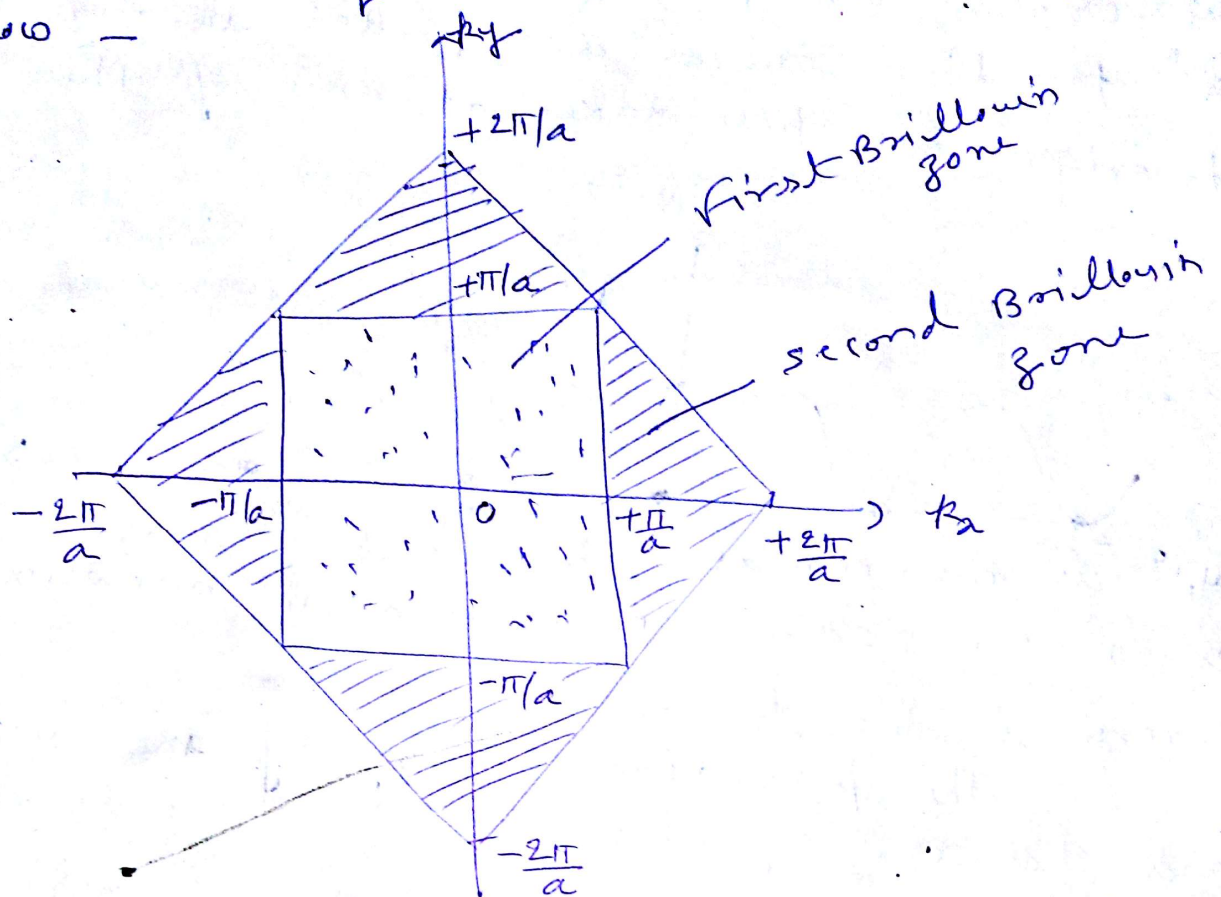
$|\psi_1|^2$ is maximum at the positive ions ($x = 0, \pm a, \pm 2a, \dots$) and min in between.

Reverse is for $|\psi_2|^2$.

The potential energy of an electron in a lattice of positive ions is greatest midway between the ions and least at the ions. So an electron for $k = \pi/a$ have two different energies E_1 and E_2 . No other solutions are possible when $k = \pi/a$, and no electron can have an energy between E_1 and E_2 . The difference $E_1 - E_2$ is the energy gap. This is the origin of all the energy gaps.

In the $\epsilon-k$ curve, the range of k values between $-\pi/a$ and $+\pi/a$ defines the first Brillouin zone, those of k values between $-2\pi/a$ and $-\pi/a$ and between $+\pi/a$ and $+\frac{2\pi}{a}$ define the second Brillouin zone. The first and second Brillouin zones of a two-

dimensional square lattice are shown below -



Effective Mass of Electron

An electron in a crystal interacts with the crystal lattice. Hence its behaviour towards external forces is ~~free~~ different from that of a free electron. The deviation of the electron behaviour can be taken by considering the electron to have an effective mass m^* in place of free-space mass m . The effective mass, m^* depends on the nature of the crystal lattice. m^* may be much larger or much smaller than m , and it may be negative.

Let an electron moving along the x -axis in a crystal in the presence of an external electric field E . The external force acting on the electron is eE . Suppose that the electron gains velocity v over a distance dx in time dt .

Then Work done = Resulting change in energy $\frac{dE}{dE}$

$$\begin{aligned} dE &= eE dx \\ &= eE v dt \quad \left[v = \frac{dx}{dt} \right] \end{aligned}$$

We know that the velocity of particle is equal to the group velocity v_g

$$dE = eE v_g dt \quad \text{--- (1)}$$

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From the Einstein de-Broglie relation

$$E = h\nu = \frac{h}{2\pi} \omega$$

where ω is the angular frequency.
Differentiating it

$$dE = \frac{h}{2\pi} d\omega = \frac{h}{2\pi} \frac{d\omega}{dk} \cdot dk \quad \text{--- (2)}$$

$$\text{But } \frac{d\omega}{dk} = v_g$$

$$\therefore dE = \frac{h}{2\pi} v_g dk$$

Comparing it with equation (1), we get

$$eE v_g dt = \frac{h}{2\pi} v_g dk$$

$$eE dt = \frac{h}{2\pi} dk$$

$$\frac{dk}{dt} = \frac{2\pi}{h} eE \quad \text{--- (3)}$$

From eq (2) v_g can be written in terms of energy E

$$v_g = \frac{d\omega}{dk} = \frac{2\pi}{h} \frac{dE}{dk}$$

Diff. w.r.t time, we get.

$$\frac{dv_g}{dt} = \frac{2\pi}{h} \frac{d^2E}{dt dk} = \frac{2\pi}{h} \frac{d^2E}{dk^2} \frac{dk}{dt}$$

$$= \frac{2\pi}{h} \frac{d^2E}{dk^2} \cdot \frac{2\pi}{h} eE \quad \text{[From eq (3)]}$$

$$= \frac{4\pi^2}{h^2} \frac{d^2E}{dk^2} eE$$

$$\text{or } \frac{dv}{dt} = \left(\frac{4\pi^2}{h^2} \frac{d^2\epsilon}{dk^2} \right) eE$$

$$a = \left\{ \left(\frac{4\pi^2}{h^2} \right) \frac{d^2\epsilon}{dk^2} \right\} eE$$

$$= \left\{ \frac{4\pi^2}{h^2} \frac{d^2\epsilon}{dk^2} \right\} F$$

$$\text{or } \frac{a}{F} = \frac{4\pi^2}{h^2} \frac{d^2\epsilon}{dk^2} \quad \text{--- (4)}$$

~~For~~ Taking $m^* = \frac{F}{a}$ as the effective mass of electron in the crystal, we have

$$\frac{1}{m^*} = \frac{4\pi^2}{h^2} \frac{d^2\epsilon}{dk^2}$$

$$\text{or } m^* = \frac{h^2}{d^2\epsilon/dk^2} \quad \text{--- (5)}$$

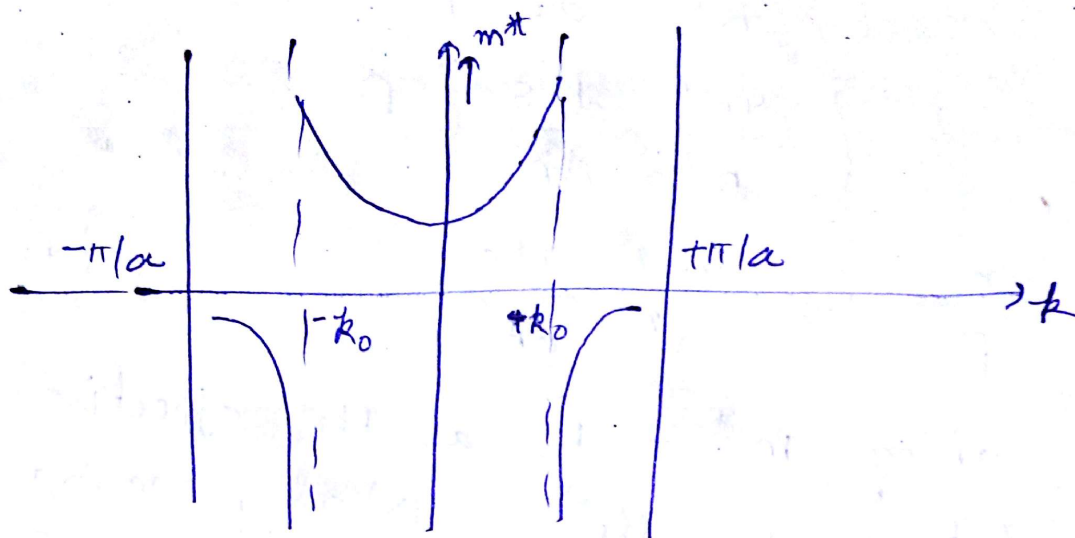
Variation of Effective Mass with k

Equation (5) represents that the effective mass depends upon $d^2\epsilon/dk^2$ and is a function of k .

From the graph $d^2\epsilon/dk^2$ is positive for values of $k = -k_0$ to $k = +k_0$ which represents the lower half of the energy band. $\frac{d^2\epsilon}{dk^2}$ is negative for values of $k = +k_0$ to $\frac{+\pi}{a}$ and $k = -k_0$ to $\frac{-\pi}{a}$, which represents the upper half of the energy band and m^*

(9)

is negative at top i.e., upper half of the energy band. At the point of inflexion of $E \rightarrow k$ curve m^* becomes infinite.



Variation of effective mass with k
A negative effective mass of electron

in the lattice physically means that the electron responds to an applied force with a decrease in momentum.

Physical significance of Effective mass:-

The significance of the effective mass is that all results of the free electron theory of metals can be taken in the more realistic band theory simply by replacing the electron mass m by the average effective mass m^* at the Fermi surface. Hence, the Fermi energy in a metal is given by $2/3$

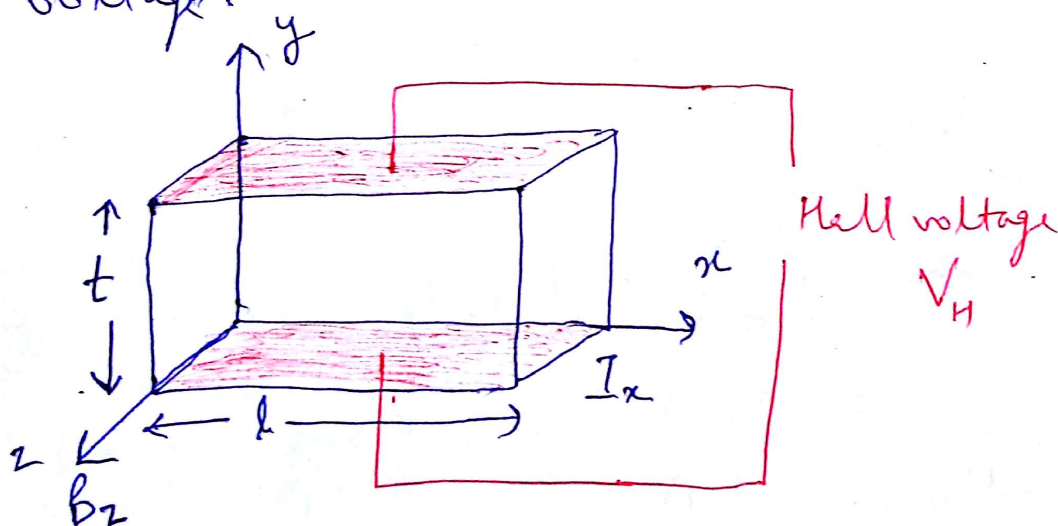
$$\epsilon_F = \frac{h^2}{2m^*} \left(\frac{3N}{8\pi V} \right)^{2/3}$$

where N/V is the density of valence electrons.

The Hall Effect :-

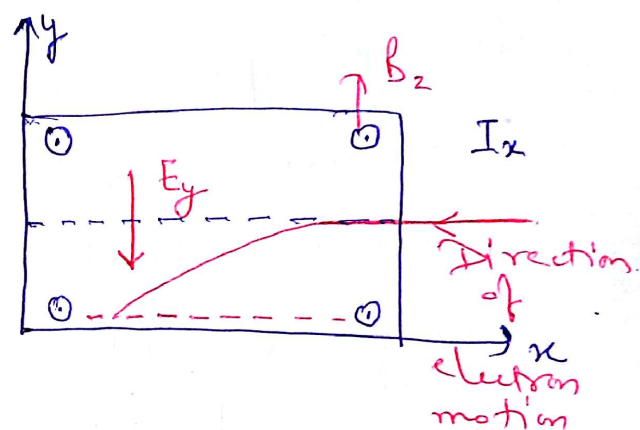
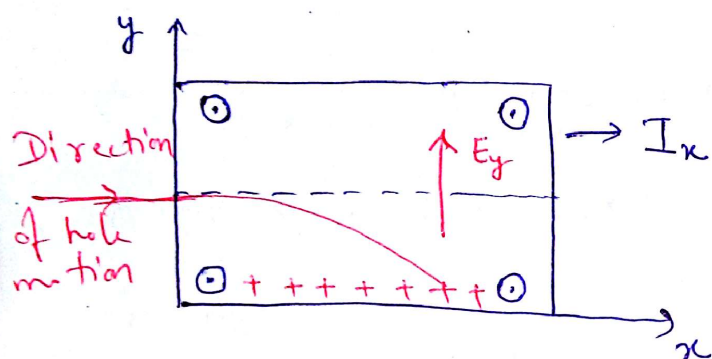
The Hall effect is the production of a potential difference (Hall voltage) across an electric conductor in which an electric current flows in the presence of a magnetic field. The Hall effect supplies the information of the sign of charge carrier.

The phenomenon of Hall effect is a consequence of motion of charge in both magnetic and electric fields. If a piece of conductor is carrying current is placed in a transverse magnetic field, then an electric field or potential difference is produced inside the conductor in a direction normal to both current and magnetic field. This phenomenon is known as Hall effect and the potential so developed is known as Hall voltage.



Consider a rectangular specimen carrying current I_x in the x -direction. A uniform magnetic field B_z along z -axis.

If conducting material is of N -type semiconductor, then current flows due to motion of electron, but for P -type, current flows due to motion of holes.



The moving charge experience an electric force (F_e) while moving in the magnetic field (applying force (F_m)).

Under equilibrium force,

$$F_e = F_m$$

$$\Rightarrow e E_y = e v_x B_z$$

$$\Rightarrow \boxed{E_y = v_x B_z} \quad \text{--- (1)}$$

If J_x is the current density

$$J_x = \frac{I_x}{A} = \frac{n e v_x A}{A} = n e v_x \quad \text{--- (2)}$$

n = density of charge carrier.

$A = l \times t = \text{length} \times \text{breadth}$ for the rectangular specimen

from eqⁿ ②, $v_x = \frac{I_x}{ne}$

Using it in eqⁿ ①,

$$\boxed{E_y = \frac{B_z I_x}{ne}} \rightarrow \text{Hall field.} \quad \text{--- ③}$$

The Hall effect is described by means of Hall coefficient R_H

$$\boxed{R_H = \frac{1}{ne}}$$

$\Rightarrow$ from eqⁿ ③

$$\boxed{R_H = \frac{E_y}{I_x B_z}}$$

for N-type $R_H = -\frac{1}{ne}$

for P-type $R_H = \frac{1}{ne}$

As R_H can be either negative or positive, so, metals can have negative or positive Hall coefficient depending upon respectively.

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Experimentally, Hall coefficient is determined by measurement of Hall voltage V_H which generates the Hall field.

Let $V_H = V_y =$ Hall voltage

$$E_y = -\frac{V_y}{t}$$

$$\begin{aligned} \therefore R_H &= -\frac{E_y}{I_x B_z} = \frac{V_y}{t} \times \frac{A}{I_x} \times \frac{1}{B_z} \\ &= \frac{V_y}{\cancel{t}} \times \frac{\cancel{lt}}{I_x} \times \frac{1}{B_z} \end{aligned}$$

$$\boxed{R_H = \frac{V_y l}{I_x B_z}}$$

Mobility and Hall angle :-

Mobility of charge carrier is defined as velocity per unit electric field.

$$\mu = \frac{v_x}{E_x}$$

If σ is conductivity of specimen material

$$I_x = \sigma E_x$$

$$\Rightarrow \mu = \frac{v_x \sigma}{I_x} = \frac{\cancel{v_x} \sigma}{n e \cancel{v_x}} = \frac{\sigma}{n e}$$

$$\Rightarrow \mu = \frac{v}{ne} = \frac{v}{\cancel{R_H} \sigma} \cdot R_H \sigma$$

$$\Rightarrow \boxed{\mu = R_H \sigma}$$

There is another important parameter, with the help of which, μ can be determined. This is called Hall angle (θ_H)

$$\tan \theta_H = \frac{E_y}{E_x}$$

$$= \frac{E_y v}{J_x} = \frac{E_y}{J_x} \cdot \frac{\mu}{R_H}$$

$$= \frac{\cancel{E_y}}{\cancel{J_x}} \times \frac{\cancel{J_x} B_z \mu}{\cancel{E_y}}$$

$$\Rightarrow \boxed{\mu = \frac{1}{B_z} \tan \theta_H}$$

Importance or Application of Hall effect :-

1. To determine the sign of Hall coefficient for a semiconductor to tell whether it is p-type or n-type
2. To determine charge carrier concentration,

$$n = \frac{1}{R_H e}$$

3. As Hall voltage is proportional to B_z and current I_x through a sample. So, the Hall effect can be used for design of magnetic flux density parameter.
4. Hall effect can be used to determine the power flow in an electromagnetic wave, because V_H is proportional to product of E_y and B_z .